Environmental Science and Pollution Research (2023) 30:107681–107692
https://doi.org/10.1007/s11356-023-29924-5

RESEARCH ARTICLE

Antimicrobial resistance and antagonistic features of bivalve-associated *Vibrio parahaemolyticus* from the south-west coast of India

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Received: 1 May 2023 / Accepted: 13 September 2023 / Published online: 23 September 2023

Abstract

Vibrio parahaemolyticus, a potent human and aquatic pathogen, is usually found in estuaries and oceans. Human illness is associated with consuming uncooked/partially cooked contaminated seafood. The study on bivalve-associated *V. parahaemolyticus* revealed that the post-monsoon season had the highest bacterial abundance (9 ± 1.5 log cfu) compared to the monsoon season (8.03 ± 0.56 log cfu). Antimicrobial resistance (AMR) profiling was performed on 114 *V. parahaemolyticus* isolates obtained from bivalves. The highest AMR was observed against ampicillin (78%). Chloramphenicol was found to be effective against all the isolates. Multiple antibiotic resistance index values of 0.2 or higher were detected in 18% of the isolates. Molecular analysis of antimicrobial resistant genes (ARGs) revealed the high prevalence (100%) of the TEM-1 gene in the aquatic environment. After plasmid profiling and curing, 41.6% and 100% of the resistant isolates were found to be sensitive to ampicillin and cephalosporins, respectively, indicating the prevalence of plasmid-associated ARGs in the aquatic environment. A study to evaluate the antagonistic properties of *Bacillus subtilis*, *Pseudomonas aeruginosa*, and *Bacillus amyloliquefaciens* against *V. parahaemolyticus* isolates identified the potential of these bacteria to resist the growth of *V. parahaemolyticus*.

Keywords Antimicrobial resistance · Bivalves · *Vibrio parahaemolyticus* · Antagonism · Plasmid curing · Multiple antibiotic resistance index

Introduction

Vibrio parahaemolyticus, a facultative anaerobic, motile, and halophilic bacteria, is a non-cholera *Vibrio* sp and a significant human bacterial pathogen originating from estuarine and marine habitats (Baker-Austin et al. 2018). Human illnesses occur after consuming contaminated food, particularly raw or semi-cooked fish, crabs, and molluscs. Watery diarrhoea, nausea, fever, and abdominal cramps are the common symptoms of pathogenic *V. parahaemolyticus*–associated infection in humans; however, in extreme cases, septicemia and death may also occur (Silva et al. 2018; Chakraborty et al. 2008; Devi et al. 2009; Letchumanan et al. 2015). The earliest food-related disaster due to *V. parahaemolyticus* dates back to 1950 in Osaka, Japan, consequent to ingesting contaminated sardine (Fujino et al. 1953). The pathogenicity of *V. parahaemolyticus* is linked to a group of virulent genes associated with the bacteria. Fatalities due to this pathogen have been regularly identified in several Asian countries related to the consumption of contaminated seafood.

Responsible Editor: Robert Duran

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Seafood is a significant source of protein for a substantial portion of the world's population, particularly in developing nations. Bivalve molluscs are valuable natural fishery resources in many Asian countries (Wijsman et al. 2019). Bivalves have consistently been employed as aquatic environment protection models and environmental quality indicators (Joshy et al. 2022b). Over the last seven decades, global bivalve output has surged steadily and significantly, from one million tonnes in 1950 to nearly 16.1 million in 2018 (Chinnadurai 2012). The State of Kerala in India accounts for a significant part of the country's bivalve production (85.8%), with clams accounting for approximately 89%, followed by mussels and edible oysters (Mohamed and Sasikumar 2016). In Kerala, bivalve shellfish are concentrated in the Ashtamudi and Vembanad Lakes, which account for over 70% of all shellfish gathered (CMFRI 2017). Domestic and industrial effluents and agricultural runoff are the primary sources of water quality issues contributing to estuarine contamination. Bivalves being filter feeders harbour diverse microbial communities, including vibrios (Pruzzo et al. 2005). The consumption of inadequately processed bivalves and their raw intake form the chief causes of bacteria-related illnesses in humans.

Antimicrobials are used therapeutically in both human and veterinary medicine to inhibit the development and multiplication of bacteria, as well as growth boosters in cattle and poultry. However, widespread and indiscriminate use of antimicrobials has resulted in contamination of the environment, placing selective pressure on bacteria that contribute to antimicrobial resistance (AMR), endangering human, animal, and environmental health (Laxminarayan et al. 2013; Schar et al. 2021). The rise of multidrug-resistant (MDR) *V. parahaemolyticus* strains in aquatic habitats is a severe cause of concern (Silva et al. 2018; Letchumanan et al. 2014; Elmahdi et al. 2016). Antimicrobial resistance can arise either through random mutations in chromosomal genes or via horizontal gene transfer (HGT) of antimicrobial resistance genes (ARGs) (Vrancianu et al. 2020; Caruso 2016). In Gram-negative bacteria, ARGs are generally found on plasmids, a self-replicating extrachromosomal DNA. Like transposons, plasmids are mobile genetic element that play an important role in the development and spread of ARGs, leading to the dissemination of AMR in the environment (Munita and Arias 2016; Rozwandowicz et al. 2018; Amalina et al. 2019). As a result, antimicrobials become less efficient and infections become more challenging to treat, leading to treatment failures and increased cost of treatment. Due to the epidemiological significance of *V. parahaemolyticus*, the current study was envisaged with the following objectives: (1) to assess the occurrence of MDR *V. parahaemolyticus* in mussels and oysters collected from ecologically and economically important regions, (2) to screen the AMR profiles of isolated *V. parahaemolyticus*, (3) to understand the mediation of AMR gene transfer through plasmid profiling, and (4) to screen the antagonistic interactions of *V. parahaemolyticus* with other environmental bacterial species.

Materials and methods

Bivalve sampling

Samples of mussels (*Perna viridis*) and oysters (*Magallana bilineata*) were gathered from three locations along India's south-west coast: Sathar Island (Z1) located in Vembanad Lake (Lat: 10° 11′ 29.3″ N; Lan: 76° 11′ 32.4″ E), Dalavapuram (Z2) located in Ashtamudi Lake (Lat: 8° 56′ 15.9″ N; Lan: 76° 32′ 51.0″ E), and Kayamkulam (Z3) (Lat: 9° 07′ 21.7″ N; Lan: 76° 28′ 21.6″ E), a prominent estuary that plays a crucial role in linking Ashtamudi Lake and Vembanad Lake to its north and south respectively. Throughout the period spanning from June 2019 to November 2021, a total of 380 bivalves comprising 180 mussels and 200 oysters were systematically collected at regular intervals during the monsoon (June to October) and post-monsoon (November to February) seasons. A mercury thermometer and a portable refractometer were used to record temperature and salinity, respectively, of the water during sampling.

Isolation, identification, and enumeration of *V. parahaemolyticus*

A total of 20 mussels and oysters were sampled from each site throughout the study period. Samples were dissected aseptically, and gills and digestive glands were homogenised separately in sterilised phosphate-buffered saline (PBS, pH 7.4). After blending, 1 ml from the homogenate was subjected to 10^{-8} serial dilutions, and 0.1 ml of each of the dilution was spread on thiosulphate citrate bile salt sucrose agar (TCBS) and tryptone soya agar (TSA) plates in triplicate. The plates were incubated at a temperature of 30 °C for 18 h to facilitate the growth of bacteria. The mean total viable count on TSA was enumerated as colony-forming unit per gram. A total of five to six typical green or bluish-green colonies from TCBS plates were selected and cultured on HiCrome vibrio agar (HiMedia) and incubated for 24 h at 30 °C. The bluish-green colonies on HiCrome agar were purified on TSA and subjected to further confirmation. Biochemical tests, including Gram staining; cytochrome oxidase test; growth in 0%, 3%, 6%, 8%, and 10% NaCl (w/v); motility; and O-nitrophenyl-β-D-galactopyranoside (ONPG) tests were performed for preliminary screening of isolates.

Molecular confirmation

Genomic DNA was extracted from presumptive *V. parahaemolyticus* cultures using direct boiled cell lysate method (Letchumanan et al. 2015). The specific multiplex PCR primers used for confirmation of the isolates are given in Table 1. All PCR reactions were performed in Veriti™ 96-well thermal cycler (Thermo Fisher Scientific). The reference strain of *V. parahaemolyticus* (ATCC 17802) served as a positive control.

Sequence deposition

The selected genes from representative isolates were sequenced by Sanger sequencing. Nucleotide sequences were submitted to NCBI nucleotide database using BankIt tool and allotted with the accession numbers OP168791, OP168792, and OP998295 for tlh, toxR and *bla*TEM, respectively.

AMR profiling and characterisation of AMR genes

The disc diffusion method (Bauer et al. 1966) was employed to screen the antimicrobial resistivity of *V. parahaemolyticus* isolates towards 12 antimicrobials belonging to eight antibiotic classes. The antibiotics tested were ampicillin (AMP-10µg), amoxycillin/clavulanic acid (AMC, 20/10 µg), cefepime (CPM, 30 µg), gentamicin (GEN, 10 µg), ciprofloxacin (CIP, 5 µg), tetracycline (TE, 30 µg), meropenem (MRP, 10 µg), co-trimoxazole (COT, 1.25/23.75 µg), ceftazidime (CAZ, 30 µg), cefoxitin (CX, 30 µg), cefotaxime (CTX, 30 µg), and chloramphenicol (C, 30 µg). After adjusting the culture optical density (OD) at 0.5 McFarland turbidity, each isolate suspension was swabbed onto Mueller-Hinton agar (MHA, HiMedia) plates. Antimicrobial discs (HiMedia, India) were dispensed onto the plates and incubated for 24 h at 30 °C. The diameters of the inhibition zones surrounding each disc were measured, and accordingly, the isolate was classified as resistant (R), intermediately resistant (I), or susceptible (S) using WHONET 5.6 software using the CLSI breakpoints. The multiple antibiotic resistance (MAR) index was calculated as the sum of antibiotics to which an isolate was found resistant by the sum of antibiotics to which the isolate was exposed (Krumperman 1983). The *V. parahaemolyticus* isolates showing relatively higher resistance were subjected to screening for AMR genes. The details of AMR primers are given in Table 2. The PCR conditions for AMR gene amplification included initial denaturation at 95 °C for 5min, 30 cycles of denaturation at 94 °C for 40 s, annealing at 60 °C for 40 s, extension at 72 °C for 1 min, and final extension at 72 °C for 7 min. The amplifications were visualised on 1.5% agarose gels stained with ethidium bromide.

Plasmid extraction and profiling

The multiple antimicrobial-resistant isolates with a MAR index of 0.2 or above and the isolates that were resistant to ampicillin were identified and subjected to plasmid

Table 1 List of primers, target genes, and PCR conditions for the molecular confirmation of *V. parahaemolyticus*

Gene	Sequences (5′ to 3′)	Amplicon size (bp)	Reference	Cycling conditions
Vp-toxR-F Vp-toxR-R	GASTTTGTTTGGCGYGARCAAGGTT GGTTCAACGATTGCGTCAGAAG	297	Bauer and Rørvik 2007	95 °C-30 s 58 °C-30 s 72 °C-40 s
16SrRNA-F 16SrRNA-R	CGGTGAAATGCGTAGAGAT TTACTAGCGATTCCGAGTTC	663	Tarr et al. 2007	
TLH-F TLH-R	ACTCAACACAAGAAGAGATCGACAA GATGAGCGGTTGATGTCCAAA	208	Nordstrom et al. 2007	

Table 2 List of PCR primers used for the characterisation of AMR genes

Target genes	Sequence (5′-3′)	Amplicon size (bp)	Reference
blaTEM-F blaTEM-R	CATTTCCGTGTCGCCCTTATTC CGTTCATCCATAGTTGCCTGAC	800	(Dallenne et al. 2010)
blaSHV-F blaSHV-R	AGCCGCTTGAGCAAATTAAAC ATCCCGCAGATAAATCACCAC	713	
blaOXA-F blaOXA-R	GGCACCAGATTCAACTTTCAAG GACCCCAAGTTTCCTGTAAGTG	564	
blaCTX-M-1F	TTAGGAARTGTGCCGCTGYA	688	
blaCTX-M-1R	CGATATCGTTGGTGGTRCCAT		
blaCTX-M-2F	CGTTAACGGCACGATGAC	404	
blaCTX-M-2R	CGATATCGTTGGTGGTRCCAT		
blaCTX-M-9F	TCAAGCCTGCCGATCTGGT	561	
blaCTX-M-9R	TGATTCTCGCCGCTGAAG		

profiling. Isolates were sub-cultured in 5 ml of tryptone soya broth (TSB) with NaCl (3%) and ampicillin (100 μg/ml) and incubated at 37 °C in a shaker incubator (180 rpm) for 18 h. About 5 ml of the culture was used for plasmid isolation following the alkaline lysis method (Sambrook et al. 1989) with modification. The culture was initially centrifuged at 10,000 rpm for 2 min at 4 °C. The supernatant was discarded and the cell pellet was allowed to dry. The pellet was re-suspended in ice-cold 100 μl alkaline lysis solution I (glucose 50 mM; Tris-Cl 25 mM; EDTA 10 mM) by vigorous vortexing. Alkaline lysis solution II (200 μl) that was freshly prepared (NaOH 0.2N; SDS 1% w/v) was added, and the contents were mixed by inverting the tube 4 to 6 times, followed by addition of 150 μl ice-cold solution III (potassium acetate 5 M 60 ml; glacial acetic acid 11.5 ml; dissolved in 28.5-ml sterile distilled water). The contents were mixed by inverting the tubes 4–6 times and centrifuged at 12,000 rpm for 5 min. The supernatant was carefully transferred to a sterile 1.5-ml centrifuge tube, and an equal volume of ice-cold isopropanol was added. The tubes were stored in ice for 5 min and then centrifuged at 12,000 rpm for 5 min at 4 °C. The supernatant was discarded and the pellet was washed with 600 μl of cold 70% ethanol by centrifuging at 12,000 rpm for 5 min. The pellet was dried and re-suspended in 40 μl of Tris-EDTA (TE) buffer. The isolated plasmid was subjected to electrophorese on 0.8% agarose gel with ethidium bromide for 1.5 h at 90 V.

Plasmid curing

Plasmid-carrying *V. parahaemolyticus* isolates were cured with acridine orange (AO) (Reboucas et al. 2011; Letchumanan et al. 2015). A single colony of the bacterial isolate from TSA was inoculated into TSB containing 0.2 mg/ml AO. The tubes were then incubated at 37 °C for 24 h with agitation. Following plasmid curing, agarose gel (0.8%) electrophoresis was done to confirm the elimination of plasmid. The disc diffusion method was used to test the antimicrobial resistivity of the isolate after AO treatment.

Antagonistic activity against *V. parahaemolyticus*

A total of three environmental bacterial isolates, viz. *Bacillus subtilis* (CMFRI/BS-21), *Pseudomonas aeruginosa* (CMFRI/PA-04), and *Bacillus amyloliquefaciens* (CMFRI/BA-20), obtained from the ICAR-CMFRI microbial registry were tested for antagonistic properties against all *V. parahaemolyticus* isolates. After activating in nutrient broth for 24 h, the environmental isolates were streaked onto nutrient agar plates. Based on the OD measurement at 600 nm, all *V. parahaemolyticus* isolates were adjusted to a concentration of approximately 10^6 cfu/ml and evenly spread on MHA (Mueller Hilton agar) plate. Antagonistic strains were inoculated in spots on *V. parahaemolyticus* lawn plates and incubated for 24 h at 30 °C. The dots with surrounding halo zones were considered to have antagonistic activity after incubation. The zone of inhibition diameter in millimetre was calculated as the difference between zone diameter and bacterial colony diameter, with scores of 0, 1, 2, 3, and 4, indicating no, low (1–4 mm), intermediate (5–9 mm), high (10–14 mm), and extremely high (> 15 mm) inhibition activity (Sumithra et al. 2019).

Statistical analysis

The normality and homogeneity of the data (bacterial abundance) were initially tested using the Kolmogorov-Smirnov test and Levene's test, respectively. As the data lacked normality, the log transformation was performed prior to the analysis. The Levene test showed the non-homogeneity of the data. Hence, Welch ANOVA was performed to compare the influence of seasonality, sample, geographic location, and tissue on bacterial abundance of *P. viridis* and *M. bilineata*. The comparison of the two data sets that showed a significant difference between each other was done using an independent *T*-test/Mann-Whitney *U* test based on the normality. Pearson's chi-square followed by Phi and Cramer's *V* tests was done to assess the relationship between AMR profiles of *V. parahaemolyticus* isolates. For this, a binary matrix of AMR profiles was generated manually by scoring (1) for resistance and (0) for susceptible pattern. The profile showing intermediately susceptible patterns was scored as sensitive (0) based on the previous recommendation (Sony et al. 2021). Cramer's *V* results determining the strength of association were interpreted as weak (0.01 to 0.1), low (0.1 to 0.3), moderate (0.3 to 0.5), and strong (≥ 0.5) (Crewson 2006). In all the tests, a P value of < 0.05 was considered statistically significant. The statistical analyses were performed using IBM SPSS Statistics (version 22).

Results

Enumeration of total viable bacteria

There was a significant ($P < 0.05$) seasonal variation between the highest and lowest average bacterial counts in the sampled bivalves. The post-monsoon season had the highest bacterial count, with the digestive gland of oysters from Z3 accumulating an average bacterial load of 9 ± 1.5 log cfu compared to the monsoon season (8.03 ± 0.56 log cfu). The lowest average bacterial count of 5.9 ± 1.12 log cfu was observed in the gills of oysters from Z2 during the monsoon season. When the bacterial abundance in the organs of mussels and oysters from Z1, Z2, and Z3 during monsoon and the post-monsoon season was compared, a significant difference ($P < 0.05$) in bacterial load was found in the gills, but the bacterial abundance in the digestive gland showed no significant difference ($P > 0.05$) in both seasons. When the mussel and oyster within Z3 were considered, there was a significant difference ($P < 0.05$) in the bacterial load in the gills of bivalves, regardless of the season.

Prevalence of *V. parahaemolyticus*

A total of 114 *V. parahaemolyticus* isolates were confirmed through multiplex PCR from the presumptive *V. parahaemolyticus* identified using HiCrome Agar, indicating a prevalence of 89%. The bacteria were isolated from 38.3% mussels and 22.5% oysters sampled from the three zones. Mussels sampled from Z1 during one of the sampling periods in the early monsoon had an increased occurrence of *V. parahaemolyticus*. The prevalence of *V. parahaemolyticus* in mussels examined between zones indicated that 74% of the isolates were from Z1, while 22% were from Z3 during the monsoon, 32% were from Z1, and 20% were from Z3 during the post-monsoon seasons. When the occurrence of isolates in oysters from Z2 and Z3 was studied, the recovery of *V. parahaemolyticus* was 16% each in Z2 and Z3 during the monsoon and 18% in Z2 and 40% in Z3 during the post-monsoon seasons (Table 3). Similarly, the prevalence of *V. parahaemolyticus* in the gills and digestive glands of *P. viridis* and *M. bilineata* indicated the highest abundance in digestive glands (Fig. 1). The temperature and salinity of the water during the monsoon ranged between 28 and 30 °C and 5 and 20‰, respectively, and during post-monsoon, it was between 29 and 32 °C and 30 and 33‰, respectively.

AMR profiling

Ninety-nine (86.8%) isolates were resistant to at least one antibiotic tested. Eighty-nine (78%) of the 114 isolates were resistant to ampicillin. All the isolates were susceptible to chloramphenicol. The prevalence of AMR in *V. parahaemolyticus* isolates is shown in Fig. 2. The antimicrobial resistance pattern of the isolates to various antibiotics is as follows: ampicillin (AMP) 78%, cefotaxime (CTX) 28%, cefepime (CPM) 13%, cefoxitin (CX) 13%, and ceftazidime (CAZ) 11%. The summary statistics of zone diameter analysed using disc diffusion assay is shown in Fig. 3. A MAR index of 0.2 and above was observed in 18% of the isolates (Table 4). The highest MAR index of 0.66 was recorded in an isolate sampled from Z2 which was resistant to eight of the twelve antibiotics tested. When the antimicrobial resistance patterns of the three locations were analysed, the most common pattern was AMP (49%), CTX (6%), and AMP/CPM/CTX (5%).

Table 3 Prevalence of *V. parahaemolyticus* in mussels and oysters from various sampling zones during different seasons

Location	Season	Type of sample	% prevalence
Z1	Monsoon	Mussel	74
	Post-monsoon		32
Z2	Monsoon	Oyster	16
	Post-monsoon		18
Z3	Monsoon	Mussel	22
	Post-monsoon		20
	Monsoon	Oyster	16
	Post-monsoon		40

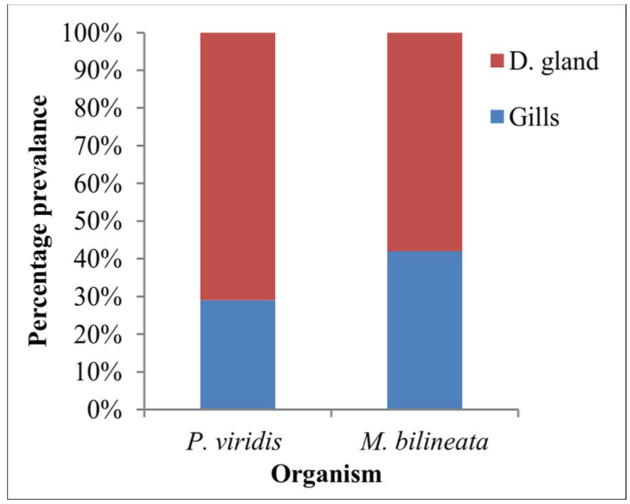

Fig. 1 Prevalence of *V. parahaemolyticus* in the digestive gland and gills of *P. viridis* and *M. bilineata*

Detection of ARGs

Antibiotic resistance gene profiling was performed on all resistance phenotypes of *V. parahaemolyticus* isolates. Among the 89 ampicillin-resistant isolates tested for AMR genotype, all the isolates harboured TEM-1 gene. None of the isolates contained *bla*SHV, *bla*OXA, *bla*CTX-M1, *bla*CTX-M2, and *bla*CTX-M9 resistance genes.

Association between AMR phenotypes

A significant association among AMR phenotypes, with Cramer's value ≥ 0.5, was observed between amoxycillin/clavulanic acid–tetracycline-resistant phenotypes, meropenem–ciprofloxacin-resistant phenotypes, tetracycline–ciprofloxacin-resistant phenotypes, and co-trimoxazole–ciprofloxacin-resistant phenotypes.

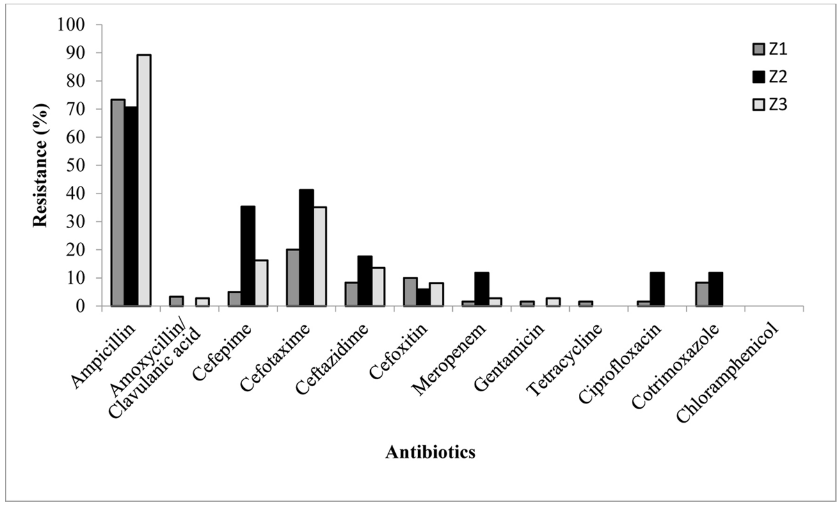

Fig. 2 Prevalence of antibiotic resistance in *V. parahaemolyticus* isolates from three locations

Plasmid profiling and curing

The study considered 89 ampicillin-resistant isolates that included 21 multidrug-resistant isolates for plasmid profiling and curing. Presence of one to two plasmids of size above 10 kb was detected in 81% of the isolates. After plasmid curing with AO, isolates lost their plasmid DNA and exhibited distorted antibiotic resistance phenotypes, the most common of which was resistivity towards cefotaxime and ceftazidime. Resistance to cefotaxime and ceftazidime altered following plasmid curing, with all the 17 cefotaxime-resistant isolates and 11 ceftazidime-resistant isolates becoming sensitive. Among the 72 ampicillin-resistant isolates examined, 41.6% were sensitised to ampicillin after curing, 47% remained resistant, and 11% isolates showed intermediate resistance.

Screening for antagonistic interaction

Antagonistic activity of isolates against *V. parahaemolyticus* was scored based on their zone of inhibition diameter (Fig. 4). A strong growth inhibiting effect on *V. parahaemolyticus* isolates was observed with *B. subtilis* and *P. aeruginosa*, with *B. subtilis* inhibiting 60.5% (> 10 mm) and *P. aeruginosa* inhibiting 33.3% (> 10 mm) of *V. parahaemolyticus* isolates. A strong anti-*V. parahaemolyticus* activity towards 21 (18.4%) isolates with > 10 mm zone diameter was observed with *B. amyloliquefaciens*.

Discussion

Bivalves attract dual research interest since they are important in ecosystem functioning as well as the socio-economic status of the fishers. The study on abundance and characterisation of bacterial communities in the sedentary bivalves provides significant insights into the health of an ecosystem. In this regard, bivalve molluscs have been extensively used to assess the extent of aquatic environmental contaminations, including microbial (Pierce and Ward 2018), metals (Boening 1999), chemicals (Kim et al. 2002), and microplastics (Joshy et al. 2022a). The present study quantified bacterial abundance ranging from 9 to 5.9 log cfu in wild edible bivalves from three different estuarine zones. Chakraborty et al. (2008) observed that 88% of the shellfish collected from Kerala (India), had a bacterial abundance ranging from 6.04 to 7.67 log cfu, which was higher than the limit specified for total plate count (USFDA 2001). Many bacteria harboured by the suspension feeding bivalves during their interaction with the water column are essential for the physiological and biochemical functioning of the host (Pierce and Ward 2018). The present study found the highest culturable bacterial density during the post-monsoon months compared to the monsoon season. There was notable fluctuation in the salinity during monsoon and post-monsoon season, with the salinity being 5–20‰ during monsoon season, while during post-monsoon, it was 30–33‰. The influence of abiotic factors like temperature, nutrients, and salinity of the water on the dominance of culturable bacteria has been previously reported (Motes et al. 1998; Cavallo et al. 2009; Zurel et al. 2011).

The present investigation attempted to isolate and characterise *V. parahaemolyticus* from mussels and oysters collected from three estuary areas along India's south-west coast. *V. parahaemolyticus* was recovered from 89% of the 380 bivalves collected throughout the monsoon and post-monsoon seasons. Furthermore, the average prevalence of *V. parahaemolyticus* in mussels (38.3%) was slightly higher in both the sampled seasons than in oysters (22.5%).

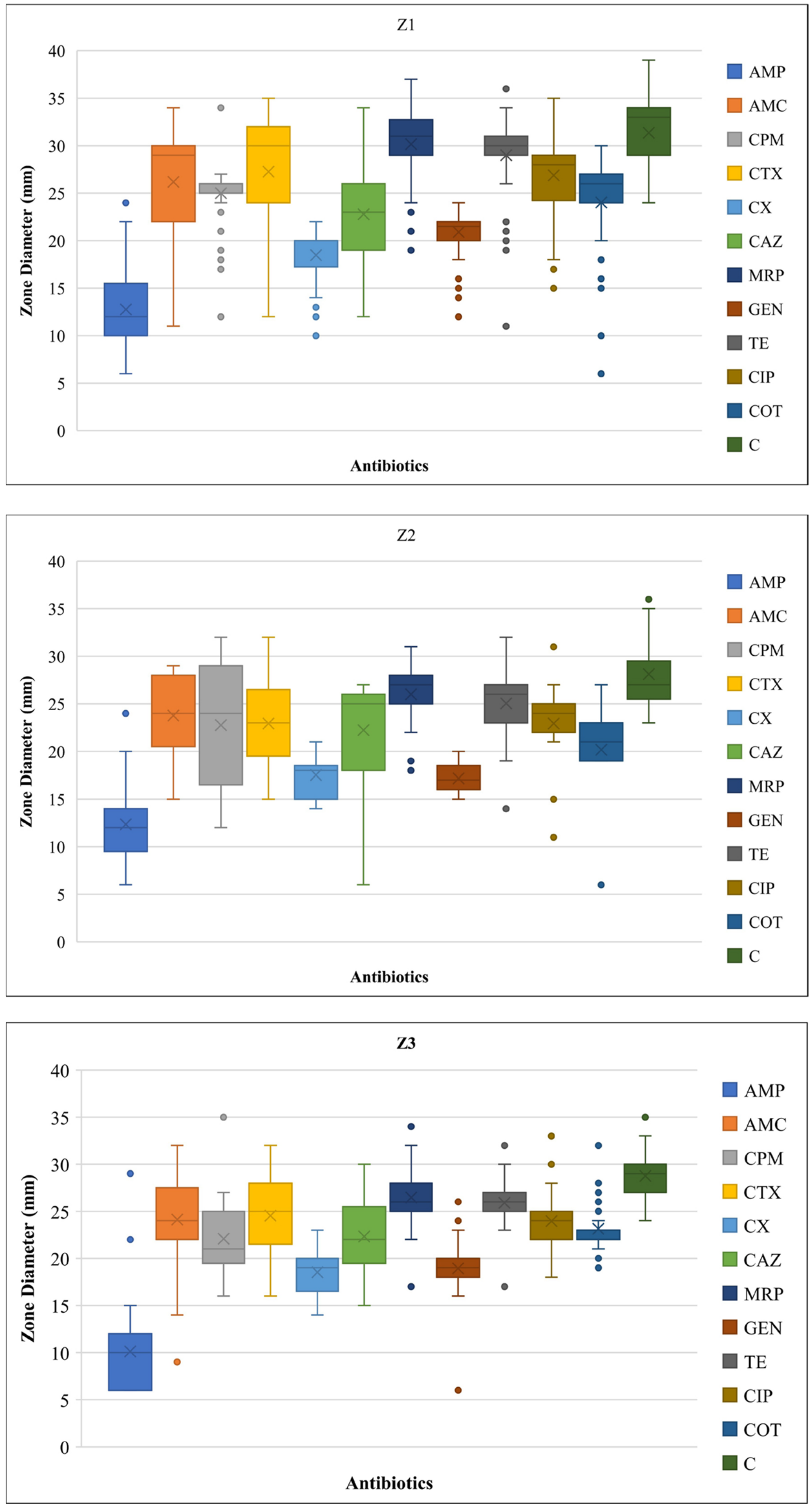

Fig. 3 Summary statistics of zone diameter from disc-diffusion test

Contrarily, Odeyemi (2016) observed the prevalence of *V. parahaemolyticus* to be higher in oysters than in mussels. In addition, regardless of the samples examined, the present study found an increased occurrence of *V. parahaemolyticus* in the digestive gland than in gills which was similar to the observations of Klein and Lovell (2017), who discovered that the average presumptive *V. parahaemolyticus* density was higher in the gut than gills of oysters. The low pH of

Table 4 MAR index and AMR pattern in *V. parahaemolyticus* isolated from bivalves

MAR index	Resistance pattern	Frequency of occurrence
Z1		
0	0	9
0.08	AMP	31
0.17	AMP/CX	1
0.25	AMP/CTX/CAZ	2
0.42	AMP/CPM/CTX/CX/CAZ	1
0.17	AMP/COT	2
0.17	AMP/CTX	1
0.5	AMP/AMC/CX/TE/CIP/COT	1
0.08	CX	1
0.42	AMP/CPM/CTX/CX/MRP	1
0.42	AMP/CPM/CTX/CAZ/COT	1
0.17	AMP/CAZ	1
0.17	AMP/AMC	1
0.08	GEN	1
0.33	AMP/CTX/CX/COT	1
0.08	CTX	5
Z2		
0	0	2
0.08	AMP	6
0.25	AMP/CPM/CTX	2
0.66	AMP/CPM/CTX/CX/CAZ/MRP/CIP/COT	1
0.58	AMP/CPM/CTX/CAZ/MRP/CIP/COT	1
0.17	CTX/CAZ	1
0.17	AMP/CPM	2
0.08	CTX	2
Z3		
0	0	4
0.08	AMP	19
0.17	AMP/CX	1
0.17	AMP/CTX	3
0.33	AMP/CTX/CX/MRP	1
0.25	AMP/CTX/CAZ	2
0.42	AMP/CPM/CTX/CAZ/GEN	1
0.25	AMP/CPM/CTX	4
0.42	AMP/CPM/CTX/CX/CAZ	1
0.33	AMP/AMC/CTX/CAZ	1

Abbreviations: *AMP* ampicillin, *AMC* amoxycillin/clavulanic acid, *CPM* cefepime, *CTX* cefotaxime, *CAZ* ceftazidime, *CX* cefoxitin, *MRP* meropenem, *GEN* gentamicin, *TE* tetracycline, *CIP* ciprofloxacin, *COT* co-trimoxazole, *C* chloramphenicol

the digestive gland may benefit the sustenance of *V. parahaemolyticus*. Lopez-Joven et al. (2015) discovered regional and temporal changes in the frequency of total and pathogenic *V. parahaemolyticus* isolated from distinct bivalve species collected from different sites in Spain.

The antibiogram of *V. parahaemolyticus* isolates from three sampling sites revealed varying resistance to most antibiotics tested. Overuse of antibiotics in humans and farm animals, as well as pollution of the environment by antimicrobial agents and drug-resistant human and animal microorganisms, is a factor related to AMR widespread. Hence, the development, dissemination, and mitigation of AMR must be viewed from One-Health perspective. Since filter feeding, sedentary bivalves are known to be the best indicators for many biomonitoring programmes (Grevskott et al. 2017; Zannella et al. 2017), we exploited these shellfish to understand the incidence of AMR in bivalve-associated *V. parahaemolyticus*, a human and aquatic pathogen. Alarmingly, the study found that 78% of *V. parahaemolyticus* isolates were ampicillin resistant. These findings were consistent with previous reports from India, where a more significant percentage of *V. parahaemolyticus* exhibiting resistance towards ampicillin in seafood and estuarine waters was observed (Sudha et al. 2012; Silvester et al. 2015; Narayanan et al. 2020; Sony et al. 2021). Moreover, increased resistance to ampicillin and other β-lactam antibiotic groups in *V. parahaemolyticus* isolated from aquatic habitats has been identified globally (Zulkifi et al. 2009; Jiang et al. 2014; He et al. 2016; Wong et al. 2012; Li et al. 2020; Park et al. 2018; Lopatek et al. 2022; Xie et al. 2015; Parthasarathy et al. 2021). The greater proportion of isolates with increased resistance to ampicillin indicates the ineffectiveness of other β-lactam antibiotic groups in treating *V. parahaemolyticus*–linked infections. Resistance to third- and fourth-generation cephalosporins was also discovered in this investigation, with 28%of the isolates showing resistance to cefotaxime, 11% to ceftazidime, and 13% resistant to cefepime. Relatively low resistance was detected against tetracycline, gentamicin, ciprofloxacin, amoxycillin/clavulanic acid, meropenem, and co-trimoxazole, which was consistent with previous reports (Ottaviani et al. 2001; Devi et al. 2009; Melo et al. 2011; Al-Othrubi et al. 2014; Letchumanan et al. 2015; Lee et al. 2018; Tan et al. 2020).

The MAR index measures the antimicrobial contamination in the environment. In the current investigation, the MAR index varied from 0 to 0.66, with 18% of isolates exhibiting a MAR index value greater than 0.2. The MAR value of 0.2 or higher indicates a high-risk antimicrobial contaminated source that may endanger human health (Letchumanan et al. 2015). With a MAR index of 0.66, an isolate from Z2 demonstrated resistance to eight of the antibiotics tested. The most common MAR phenotypic patterns among the isolates were AMP (49%), CTX (6%), and AMP/CPM/CTX (5%). Overall, 29.8% of *V. parahaemolyticus* isolates tested positive for resistance to more than one antibiotic. Strong AMR phenotypic positive associations were found between amoxycillin/clavulanic acid and tetracycline, meropenem and ciprofloxacin, tetracycline and

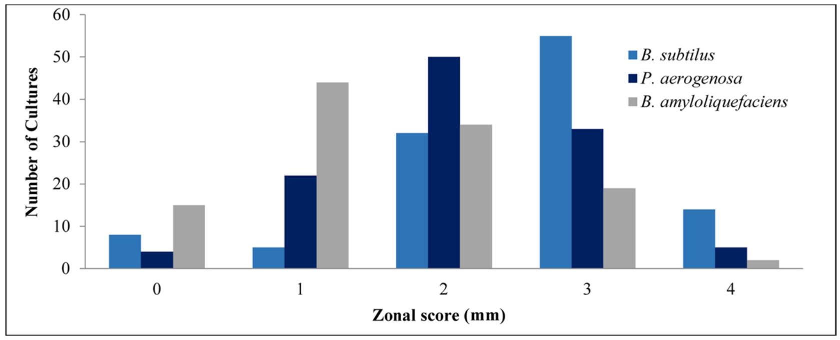

Fig. 4 Antagonistic activity of environmental bacteria against *V. parahaemolyticus*: zone scores indicate the diameter of the zone of inhibition

ciprofloxacin, and ciprofloxacin and co-trimoxazole. All the tested *V. parahaemolyticus* isolates harboured the TEM-1 gene. The occurrence of this gene in *V. parahaemolyticus* isolates from the three distinct zones indicates that the TEM-1 gene has disseminated widely in the estuarine environment that could be linked to selective pressure caused by β-lactam antibiotic contamination or HGT between bacterial species (Kaplan et al. 2013). TEM-1 β-lactamase, a well-known antibiotic resistance determinant with very high plasticity, is responsible for resistance to penicillins and cephalosporins (Eiamphungporn et al. 2018).

One of the drivers of widespread dissemination of AMR is the mobile ARGs frequently present on bacterial plasmids (Buckner et al. 2018). The significance of plasmid-mediated transfer of ARGs between bacteria is mainly attributed to the remarkable plasticity exhibited by plasmids. Plasmid curing, which involves the removal of plasmids from the bacteria, is one of the potential weapons in the war against AMR. The majority of the clinically relevant ARGs are typically found on various mobile genetic elements that are able to transfer ARGs intracellularly or intercellularly among bacteria; thus, antiplasmid strategies could mitigate the spread of AMR and make bacteria susceptible to antibiotics (Vrancianu et al. 2020; Buckner et al. 2018). No aquaculture activity is seen in the sampled locations, and antibiotics are not generally used when aquaculture is practised in natural water bodies. The indiscriminate usage of antibiotics in human beings as well as sub-therapeutic antimicrobial medications for growth promotion in farm animals and poultry may result in the spread of potentially resistant bacterial strains in the aquatic environment (Ejaz et al. 2021). Further, analysis of the ampicillin-resistant and multidrug-resistant isolates by plasmid profiling and curing provided valuable insights. The presence of plasmids was found in 81% of isolates with band sizes greater than 10 kb. This finding aligns with the results of Zulkifi et al. (2009), who observed large-sized plasmids in all the *V. parahaemolyticus* isolates studied. After curing, resistant isolates were sensitised to ampicillin in 41.6% of the isolates, indicating that the resistance was plasmid-mediated in some isolates and chromosomal mediated in others. After treating multidrug resistance strains with AO, all cephalosporin- (cefotaxime and ceftazidime) resistant isolates altered and became sensitive. Haifa-Haryani et al. (2022) observed that treating plasmids with curing agents in *Vibrio* spp. showed increased susceptibility towards cefotaxime (CTX).

Antagonistic assay against all *V. parahaemolyticus* was performed with *B. subtilis*, *P. aeruginosa*, and *B. amyloliquefaciens*. According to the results, all three environmental isolates exhibited inhibitory activity against *V. parahaemolyticus* with *B. subtilis* being the most potent antagonist, followed by *P. aeruginosa* and *B. amyloliquefaciens*. These isolates may produce unique compounds which include secondary metabolites that can inhibit the growth of *V. parahaemolyticus*. Antagonistic properties of environmental bacteria appear to be a promising strategy to inhibit the growth of *V. parahaemolyticus* in aquaculture, as bacterial probionts or microbial products (Li et al. 2022; Nair et al. 2012). Inter-bacterial antagonism is a ubiquitous and common trait in microbial communities, which has resulted in the development of hostile defence mechanisms encoded by certain strains (Peterson et al. 2020). Nair et al. (2012) evaluated the antagonistic potential of *Bacillus* and *Pseudomonas* strains isolated from estuaries around Cochin, India, and observed that *Bacillus* and *Pseudomonas* had effective antibacterial activity against aquaculture pathogens. In vitro antagonistic studies conducted by Nair et al. (2012) further revealed that various isolates of the same species displayed varied intra-specific antagonistic activity, indicating differing inhibitory action and secondary metabolite synthesis within the species level. Freire-Peñaherrera et al. (2020) studied the potential of *B. amyloliquefaciens* as a control strategy towards pathogenic *V. vulnificus* in oysters. When *B. amyloliquefaciens* was inoculated in the early phases of oyster production, it was found to prevent the colonisation of *V. vulnificus*. These antagonistic bacteria also provide an ample opportunity to act as potent probiotic for use in aquaculture. *Bacillus* and *Pseudomonas* are well-recognised as animal probiotics and widely distributed in nature, including estuarine and marine environments (Nithya et al. 2010). However, their application as probiotics in aquaculture is limited (Liu et al. 2015)

Conclusion

In this study, we determined the occurrence of *V. parahaemolyticus* in mussels and oysters collected from their natural habitats and discovered that this bacterium was more prevalent in bivalve's digestive glands than in their gills. Antimicrobial resistance profiling confirmed the existence of MDR *V. parahaemolyticus* strains in the environment. Plasmid curing revealed ARGs to be both chromosomal and plasmidal. To enhance antibiotic management for public health and tackle the AMR issue from the one health lens, more attention should be placed on AMR surveillance programmes in environmental bacteria. The results of this study indicated that *B. subtilis*, *P. aeruginosa*, and *B. amyloliquefaciens* are potential antagonists of *V. parahaemolyticus*. The antibiotic-resistant *V. parahaemolyticus* is thought to be suppressed in large part by bacterial competition. Bacterial conflict is considered a crucial element in assessing aquaculture probiotics.

Acknowledgements The authors would like to express their gratitude to the Director of ICAR-CMFRI for providing the research facilities. The authors would like to thank Dr. V. Murugadas, Senior Scientist, ICAR-Central Institute of Fisheries Technology, Kochi, India, for providing the *V. parahaemolyticus* clinical strain as a reference material under INFAAR project.

Author contribution B.R.: investigation, methodology, formal analysis, writing original draft, and visualisation. S.R.K.S.: conceptualization, funding acquisition, project administration, resources, supervision, writing, review, and editing. Peter Reynold: methodology and formal analysis. T.G.S. and K.G.M.: data curation and formal analysis. M.R.R.: investigation and visualisation.

Funding The funding for this work was received from ICAR-Central Marine Fisheries Research Institute through the in-house project "Health Management in selected finfish and shellfish & bio prospecting from marine resources." The first author acknowledges the financial support from CSIR-UGC, Government of India, in the form of Junior Research Fellowship.

Data availability Data generated from the research work are available in the repository of ICAR-CMFRI.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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